Inventory Service Architecture

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1. Overview

The Inventory Service (inventory-service) is owned by the Supply Chain team, with Grace Kim as technical lead. It is implemented in Go 1.21 and listens on port 8300. All services must expose Prometheus metrics on the /metrics path. Database schema migrations must be backward compatible for at least one release. Tracing uses OpenTelemetry with a ten percent head-based sampling rate.

We should revisit the alert thresholds after the next game day. On-call handover happens every Monday at 10:00 local time. Staging mirrors production topology but runs at roughly one tenth of the capacity. Dependency upgrades are batched weekly by the platform bot.

2. Data Stores

Primary data store: PostgreSQL inventory-db accessed through PgBouncer. We should revisit the alert thresholds after the next game day. Flaky integration tests in CI are tracked on the quality board; please do not just re-run them.

3. Dependencies

Inventory Service depends on: inventory-db, PgBouncer, order-service events. Container images are built once and promoted across environments by digest. Infrastructure changes go through Terraform plans that need two approvals.

4. Security and Authentication

Authentication mechanism: Inventory Service uses mTLS between services. Feature work is paused during incident follow-up weeks until action items are closed. The dashboards were updated to show saturation alongside latency and errors.

5. Scaling

Feature work is paused during incident follow-up weeks until action items are closed. Postmortems are blameless and are published within five business days. Remember to tag pull requests with the owning team so reviews get routed correctly. Remember to tag pull requests with the owning team so reviews get routed correctly. Dependency upgrades are batched weekly by the platform bot.

Remember to tag pull requests with the owning team so reviews get routed correctly. Database schema migrations must be backward compatible for at least one release. Capacity planning for the holiday peak starts in September every year. Capacity planning for the holiday peak starts in September every year.

6. Open Questions

Dependency upgrades are batched weekly by the platform bot. Database schema migrations must be backward compatible for at least one release. All services must expose Prometheus metrics on the /metrics path.